RACE
Regular Analysis Through Continuous Exercise

The d and f block Elements : RACE-01

Variable Oxidation States, Stability of higher O.S.

1. Sum of highest stable oxidation states of following elements is:
Sc, Zn, Ti, Mn, Cr

CDF137

Some Important Compounds of Transition Elements and their (Physical and Chemical) properties ($K_2Cr_2O_7$, K_2CrO_4, $KMnO_4$, K_2MnO_4)

2. K_2MnO_4 can be converted into $KMnO_4$ by:
(A) Passing CO_2 gas (B) By passing Cl_2
(C) Electrolytic oxidation (D) All of these

CDF128

3. In which of the following oxo-anion, all M–O bond length are not identical
(A) $MnO_4^{\ominus}$ (B) $MnO_4^{2\ominus}$ (C) $Cr_2O_7^{2\ominus}$ (D) $CrO_4^{2\ominus}$

CDF130

4. Potassium permanganate acts as an oxidant in neutral, alkaline as well as acidic medium. The final products obtained from it in the three conditions are, respectively
(A) $MnO_4^{2\ominus}$, $Mn^{3\oplus}$ and $Mn^{2\oplus}$ (B) MnO_2, $MnO_4^{2\ominus}$ and $Mn^{2\oplus}$
(C) MnO_2, $MnO_2^{\oplus}$ and $Mn^{3\oplus}$ (D) MnO_2, MnO_2 and $Mn^{3\oplus}$

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5. $K_2Cr_2O_7$ when reacts with cold and conc. H_2SO_4 gives red crystal of:
(A) $CrO_4^{2\ominus}$ (B) CrO_3 (C) $Cr_2(SO_4)_3$ (D) Cr_2O_3

CDF133

6. Which of the following statements are correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
(A) Deep red vapours are liberated
(B) Deep red vapours dissolve in NaOH forming a yellow solution.
(C) Greenish yellow gas is liberated
(D) Deep red vapours get hydrolysed in water

CDF171

Paragraph for Q. 7 and 8

MnO_2 is the most important oxide of maganese. It occurs naturally as the black coloured mineral pyrolusite. It is an oxidising agent, and decomposes to Mn_3O_4 on heating to 530°C. It is used in the preparation of potassium permanganate and in the production of Cl_2 gas. Over half a million tonnes per year of MnO_2 is used in dry batteries.

7. When MnO_2 is fused with KOH in the presence at air, the product formed is:
(A) purple colour $KMnO_4$ (B) green colour K_2MnO_4
(C) colourless $MnO_4^{\ominus}$ (D) purple colour K_2MnO_4

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8. In which of the following species, the colour is due to charge transfer.

(I) $[Mn(OH)_4]^{2\ominus}$ (II) K_2MnO_4 (III) MnO_2 (IV) $KMnO_4$

(A) I, II, III (B) II, IV (C) I, II (D) Only IV

CDF172

The Lanthanoids (Electronic Configuration, Atomic and Ionic Sizes, Oxidation States)

9. Which of the following lanthanoid has one electron in 6d subshell?

(A) La (B) Ce (C) Gd (D) None of these

CDF145

General characteristics of Lanthanoids, Chemical Reactions of The Lanthanoids

10. CeO_2 is:

(A) A good oxidising agent (B) Diamagnetic in nature

(C) Colourless compound (D) All of these

CDF139

11. Which of the following ion has maximum complex forming tendency?

(A) $La^{3\oplus}$ (B) $Ce^{3\oplus}$ (C) $Eu^{3\oplus}$ (D) $Lu^{3\oplus}$

CDF141

12. Which one of the following statements is not correct?

(A) $La(OH)_3$ is less basic than $Lu(OH)_3$.

(B) In lanthanide series, ionic radius of $Ln^{3\oplus}$ ions decrease.

(C) La is actually an element of transition series rather than lanthanide series.

(D) Atomic radii of Zr and Hf are same because of lanthanide contraction.

CDF142

The Actinoids (Electronic Configuration, Atomic and Ionic Sizes, Oxidation States)

13. Which of the following show highest oxidation state?

(A) Cl (B) Mn (C) Np (D) All of these

CDF140

14. Which of the following statement is not correct?

(A) $Lu^{3\oplus}$ has the strongest tendency toward complex formation among trivalent lanthanoid ions.

(B) Ce has maximum composition in misch metal.

(C) f-block elements can have electrons from f^0 to f^{14}.

(D) Nd, Np and Nb all are f-block elements.

CDF144

General characteristics and comparison with Lanthanoids

15. Which of the following characteristic is not the point of resemblance between lanthanoids and actinoids?

(A) Reducing property (B) Oxidation state of +3

(C) Trends of ionic radii for $M^{3\oplus}$ ions (D) Radioactivity

CDF143

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	22	D	C	B	B	ABD	B	B	D	D
Question	11	12	13	14	15					
Answer	D	A	D	D	D					

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